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Paleomagnetism of the Thetford Mines Ophiolites, Quebec

Maurice K.-SEGUIN

Département de Géologie, Université Laval, Québec, Canada

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One hundred and twenty-five (125) oriented samples (286 specimens) were obtained from thirty-three (33) sites in metavolcanic rocks and diabase dyke swarms from the ophiolite complex of Thetford Mines in the Appalachians of Southern Quebec (longitude: 71°00′-71°45′W, latitude: 45°45′-46°15′N). The metavolcanic rocks are massive or pillowed (mainly) lavas of andesitic and mostly basaltic composition and the dyke swarms are composed of basaltic and doleritic diabases; both lithological groups are metamorphosed to the subgreenschist (*pumpellyite*) facies. Pyroclastics, tuffs and volcanic agglomerates are also present in minor quantities.

The NRM intensities of the metalavas and diabases are very weak. Magnetic, titanomagnetite and occasional grains of hematite are the magnetic memory carriers. Native iron was also detected in minute quantities. In order to obtain some pertinent information relative to the stability of the NRM component, stepwise alternating field (AF) demagnetization was conducted on 30% of the specimens in the 50-800 Oersteds range and the others were demagnetized at an optimum AF intensity. Approximately 10% of the specimens underwent thermal treatment in the 100-550°C range; thermal demagnetization was judged less efficient than AF demagnetization and for this reason, the remaining specimens were all AF demagnetized.

After AF treatment, the paleopole position of the tilted ophiolite complex is 206° to 09°S (d_m=16°, d_p=10°, K=61), i.e. 26°E, 09°N (normal polarity).

Petrological and magnetic studies of the ophiolite complex of Thetford Mines indicate that the basaltic and doleritic diabase dykes which feed the overlying pillow lavas in the Thetford Mines complex have NRM intensities and Koenigsberger ratios which are comparable to those of the layer of pillow lavas. As the NRM intensities of both the pillow lavas and the diabase dyke swarms are low, they ought to be relatively unimportant contributors to magnetic anomalies. The large NRM intensities and Koenigsberger ratios of the gabbroic layer and particularly the harzburgite, dunite, wehrlite and pyroxenite of Thetford Mines suggest that the ultramafic layers are the most significant magnetic sources which might well explain the younger marine anomalies. Surface alteration and low temperature oxidation may, however, be partly responsible for the low NRM intensities of the relatively thin (1 km-thick) layer of pillow basalts of this ancient sea floor.

The absence of a reliable model of the source(s) of the magnetic lineations

is a serious handicap in the solution of numerous geological and geophysical problems such as dyke intrusion models of seafloor spreading geomagnetic reversal time-scale and paleogeographic reconstruction (BANERJEE *et al.*, 1974). This study is concentrated on these two last aspects.

1. Introduction

The paleomagnetism of ophiolitic pillow lavas is interesting both for its own properties and its relevance to the hypothesis of an oceanic origin. The interpretation of marine magnetic anomalies in terms of geomagnetic field reversals and seafloor spreading has become one of the cornerstones of global plate tectonics. However, an accurate model of the source(s) of the oceanic linear magnetic anomalies is not yet available. The answer to this problem may be found in old remnants of the oceanic crust such as the Thetford Mines ophiolite complex. In a previous paper (SEGUIN and LAURENT, 1975), a description of the structural and petrological characteristics as well as the paleomagnetic properties of the ophiolitic pillow lavas of Thetford Mines was presented. A comparison of their characteristics with their analogs from the sea floor and a discussion of the origin and evolution of features observed was also given.

2. Geology

The ophiolite complex of Thetford Mines is part of a discontinuous belt of peridotite bodies and associated rocks of the Appalachians of Southern Quebec. They constitute a string some 300 km long, called the "Serpentine Zone" which extends from Vermont to the south to Maine in the north and going through Orford, Asbestos, Thetford Mines, East Broughton, St-Fabien-de-Panet and St-Omer in the province of Quebec.

The ophiolite sequence consists of a basal amphibolite underlain by greywackes and lava flows of the Cambrian Caldwell Group and overlain by harzburgite, dunite, wehrlite, pyroxenite, gabbro, diabase dykes and pillow lavas (volcanics and volcanic breccias). This Cambro-Ordovician sequence is overlain by a phyllitic mélange of the St-Daniel Formation (Early Ordovician) and finally a Middle Ordovician flysch of the Magog Group.

Various aspects of the geology of the Thetford Mines ophiolite suite were described in the past. Early geological mapping was done by KNOX (1916) and Harvie (Thetford map-area, *Geol. Surv. Can.*, Unpublished report, 1923). More recent contributions were made by COOKE (1937, 1950), RIORDON (1953, 1954, 1957), ST-JULIEN (1970, 1975) and DEROSIER (1971). KAČIRA (1971) demon-

strated the mantle origin of the Asbestos peridotites, and LAMARCHE (1972, 1973) and LAURENT (1973, 1975) recognized the ophiolitic nature of the complex.

For the sake of clarity, a short description of the diabase dykes and the metavolcanics will be presented. The pillowed metabasalts, whose cumulative thickness reaches locally a maximum of about 600 m, make up the upper part of the Thetford Mines ophiolite. The pillow lavas contain olivine which is chloritized or silicified, pyroxene which is replaced by albite, actinolite and small amounts of epidote.

Composite dyke swarms and sills are numerous in the pillow lavas horizons and the underlying layered gabbros; they are composed of diabase quartz diorite, spilitic keratophyre and granitic differentiates. The composition of the diabase dykes is quite akin to the one of the pillow lavas.

3. Age

The pillow lavas which are located stratigraphically on top of the ophiolitic sequence yielded a radiogenic (K/Ar) age of approximately 550 m.y. (LAURENT and VALLERAND, 1974). The ophiolites have a *Middle Cambrian* age.

4. Sampling Procedure

Some 136 oriented samples (317 specimens) were collected on 36 different sites; the orientation was done with a Brunton compass. The number of samples per site varied between 2 and 9 and the number of specimens per sample between 2 and 6. The NRM component was found to be unstable at 3 sites, so that the number of samples was reduced to 125 and the number of specimens to 286. The samples were drilled on the field with a portable diamond drill. In this extremely detailed survey, efforts were made to select sites where there was an excellent exposure and where the structural geology, even though quite complex, was best understood. Great care was also taken to select fresh material. The sample locations were chosen by reference to detailed geological maps published by Cooke (1937) at 1 mile to the inch.

5. Measurements

The direction and intensity of remanent magnetization were measured with a digital spinner magnetometer (model DSM-1) manufactured by Schonstedt Inst. Co. (sensitivity $=10^{-8}$ cgs emu). Alternating field demagnetization was carried out to remove unwanted secondary components using a demagnetizer built at the University of Laval (maximum intensity of 1,800 Oersteds rms) the

performance of which was largely improved by adding 3 large concentric mu-metal cylinders around the solenoid. Thermal treatment of the specimens was done in a field-free specimen demagnetizer (model TSD-1) manufactured by Schonstedt Inst. Co.

6. Initial NRM Intensities and Directions

The NRM intensities range from 1.5×10^{-7} to 1.4×10^{-2} emu/cc. The high intensity values are more frequently found in gabbroic specimens which have an average intensity of 5×10^{-5} emu/cc. The average intensity of the pillow lavas is 3×10^{-7} emu/cc which is indeed very low. There is very little or no significant difference between the NRM intensities of the pillow lavas and the diabase dykes. The mean NRM direction of the pillow lavas *in situ* is 077°, 64° and the direction of the diabase dykes 071°, −59°. For both lithological groups, the orientation is different from the actual declination and inclination of the Earth's field (342°, 70°); this indicates that a large fraction of the remanence is of ancient origin. After tilting both groups to the paleohorizontal, the mean NRM directions are 183°, 19° and 118°, 67°. The increase in $\alpha 95$ and the decrease of K (dispersion) for the mean NRM direction of the tilted pillow lavas and diabase dykes relative to the one of the same groups *in situ* suggest that a fraction of the NRM component is post-folding.

7. AF Demagnetization

A minimum of 3 specimens (usually 4 to 8) from each site was demagnetized in steps of 50 Oersteds, from 50 to 500 Oersteds, and in steps of 100 Oersteds, from 500 to 800 Oersteds in the absence of an ambient field. About 30% of the specimens had very low NRM intensities and could be demagnetized up to 350 Oersteds only. Guided by the stepwise changes in orientation and intensity of the pilot specimens from a site, 1 and occasionally 2 AF intensities of AF demagnetization were selected using an "objective end point" stability method in which the remaining specimens from the samples of the site were AF demagnetized and their residual remanence measured. Pilot tests show that the AF intensity at which the primary thermoremanent and viscous remanent components (DUNLOP and HALE, 1977) is best isolated is located in the 150–300 Oersteds range or the 450–700 Oersteds range (Fig. 1). Normalized demagnetization intensity curves for the same test specimens from different sites are shown in Fig. 2.

The behaviour of the orientation and intensity of the residual remanence is quite complicated and difficult to interpret (Figs. 1 and 2). A statistical study

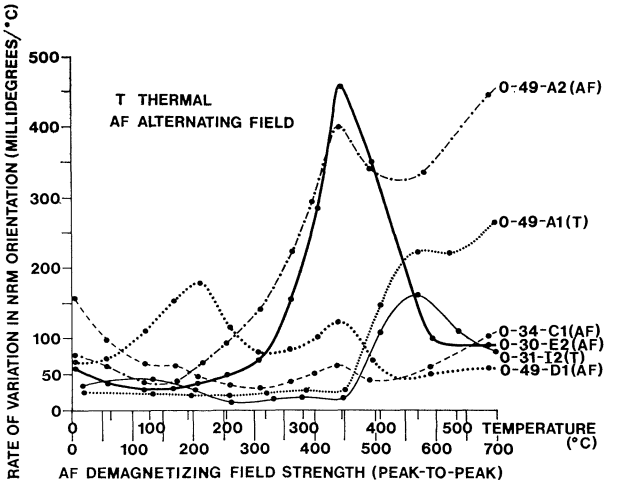


Fig. 1. Stability index plot for 6 specimens of the ophiolite complex.

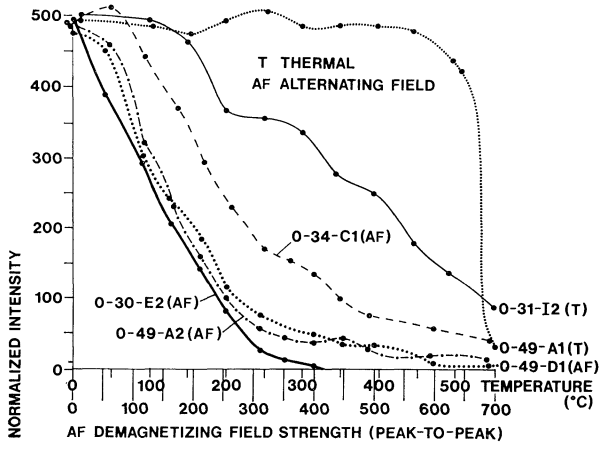


Fig. 2. Normalized remanent intensity of 6 typical specimens of the ophiolite complex as a function of the demagnetizing field strength.

(IRVING, 1964) of the orientations of the residual remanent components was carried out for both intensity ranges. 45% of the pilot specimens used indicated a relatively stable NRM component in the 150–300 Oersteds range, 30% in the 450–700 Oersteds range and the remaining 25% indicated a relatively stable NRM component in both the 150–300 and 450–700 Oersteds ranges. This statistical study was done for both pillow lavas and diabase dykes groups in their actual structural setting and once tilted to their presumed original position

Table 1. N.R.M. Study: Thetford mines ophiolite complex.

	Site No.	No. samples	No. specimens	Without demagnetization						After demagnetization						
				In situ			Corrected for tilt			*In situ*			Corrected for tilt			
				D_1 (°)	I_1 (°)	$\alpha 95$ (°)	D_2 (°)	I_2 (°)	$\alpha 95$ (°)	D_1 (°)	I_1 (°)	$\alpha 95$ (°)	Field intensity (Oersteds)	D_2 (°)	I_2 (°)	$\alpha 95$ (°)
				Part I-a) Pillow lavas and gabbros.												
	14	6	12	327	−02	22	142	−20	17	197	11	06	300	224	27	06
	15	3	6	024	52	18	105	14	15	086	53	32	150	108	−14	32
	16	4	9	315	20	21	290	−01	18	025	23	28	250	085	34	29
	17	3	7	036	34	14	043	−40	14	166	41	22	450	245	38	24
	30	6	13	204	28	24	226	03	23	179	40	16	150	171	−48	15
	31	7	15	317	63	10	311	−23	11	163	61	21	300	152	−19	23
	34	3	7	065	71	16	349	23	12	047	24	23	200	055	−31	27
	43	6	13	083	71	21	307	29	22	025	14	33	250	006	12	31
	47	4	9	263	06	16	236	22	16	074	40	11	600	012	−46	20
	49	2	7	193	34	14	250	35	11	076	62	23	200	024	25	10
Total	10	44	98	317	76	15	294	06	13	108	64	12		076	−17	18
and mean:	Paleopole position:			257°E @ 59°N			179°E @ 19°N			345°E @ 33°N				034°E @ 03°N (d_m=19°, d_p=8°, K=22)		
				Part I-b) Sheeted dykes.												
	14	2	5	341	−07	05	356	43	25	139	16	28	300	108	−60	28
	15	4	11	358	51	15	338	−17	14	205	62	19	150	274	12	18
	16	2	4	310	40	31	219	08	08	241	46	35	250	177	15	35
	17	2	5	021	26	09	043	−60	14	333	41	33	450	326	−63	33
	31	2	6	034	31	10	043	−58	11	338	24	21	300	326	−38	20
	43	2	5	066	23	18	040	13	15	—	—	—	250	—	—	—
	49	2	4	169	34	14	253	55	14	189	−02	27	200	208	30	19
Total	7	16	40	017	42	10	352	−12	17	215	60	19		261	−11	22
and mean:	Paleopole position:			071°E @ 64°N			117°E @ 37°N			262°E @ 02°N				202°E @ 34°S (d_m=22°, d_p=11°, K=16)		

							Part I-a) and b) combined.									
Total and mean:	17	60	138	356	71	08	315	02	11	145	82	19		258	−04	27
	Paleopole position:			277°E @ 69°N			164°E @ 30°N			298°E @ 31°N				204°E @ 04°S (d_m=27°, d_p=14°, K=28)		
							Part II-Pillow lavas and gabbros only.									
	18	3	7	019	72	18	122	04	17	338	53	27	300	124	30	26
	19	2	3	342	75	33	128	27	34	043	08	28	450	053	04	29
	20	3	6	244	66	29	164	17	30	319	31	32	400	144	61	33
	21	2	5	011	69	06	146	08	05	347	65	29	350	152	15	28
	22	4	8	065	55	15	093	−28	15	052	10	30	700	038	−32	29
	23	3	7	334	39	25	056	69	26	331	37	32	250	052	65	31
	24	3	6	179	44	26	153	−01	26	168	12	25	150	177	−26	26
	25	3	6	322	−28	31	324	46	32	136	22	36	200	136	−53	35
	26	3	6	346	61	22	331	−07	22	344	44	28	350	338	−22	27
	27	2	4	090	37	25	083	−34	26	106	−11	23	250	152	−41	19
	28	2	5	084	33	29	344	70	30	336	−32	29	500	031	−50	28
	29	4	8	115	63	15	270	45	15	284	21	32	450	286	−49	30
	32	2	4	083	28	08	016	46	08	083	24	05	300	021	47	04
	33	2	4	239	50	25	332	−01	32	227	48	31	500	331	−33	33
	35	2	5	281	−24	32	200	−64	34	343	33	28	250	341	−29	29
	36	3	7	207	−11	19	120	−47	19	203	−64	24	300	077	−08	25
	37	2	4	039	29	36	028	−40	36	026	−04	35	500	348	−28	35
	41	3	6	294	03	26	198	70	26	323	02	29	150	076	77	28
	42	5	10	094	58	24	349	09	23	204	08	23	200	255	29	27
	38	2	6	004	28	15	352	−18	15	009	−21	28	400	040	−04	28
	39	1	2	320	67	19	084	22	20	278	75	23	300	107	15	23
	40	3	8	303	10	18	165	72	19	291	−07	17	200	229	69	20
	44	4	8	257	21	35	246	−31	35	110	06	24	150	018	59	25
	45	2	5	197	40	25	190	−30	27	120	10	30	300	083	−58	31
	46	3	6	271	49	29	197	30	29	342	55	19	600	147	51	19
	48	3	6	090	−13	34	051	−31	33	161	−10	27	500	244	−80	29
Total and mean:	48	71	154	046	86	16	114	29	17	030	58	16		072	06	17
	Paleopole position:			297°E @ 51°N			351°E @ 04°S			027°E @ 66°N				030°E @ 14°N (d_m=18°, d_p=09°, K=57)		
	Parts I and II Combined						354°E @ 10°S							032°E @ 9°N		

(Table 1). After demagnetization, the inclination of the site mean directions of the tilted formations displaced from the southern towards the northern hemisphere (see paleopole positions of Parts I and II combined on Table 1). About 30% of the pilot specimens depict an intensity increase of the residual component in the 50–700 Oersteds range (commonly 50–100 Oersteds) and this increase is occasionally accompanied by a polarity inversion (Fig. 2). The decrease in intensity of the residual remanence is quite rapid in the 50–350 Oersteds range; the median destructive field occurs in the 25–300 Oersteds range (mean 130 Oersteds). There are notable exceptions for the pilot samples showing an intensity increase of the residual component; in this case, the median destructive field is larger than 600 Oersteds. The significance and stability of the remanent component of the pillow lavas and diabase dykes of the ophiolites are demonstrated by the departure of the orientations of the AF cleaned residual component from the recent Earth's field direction and the evidence for reversal(s).

After AF demagnetization and tilting of the formation, the paleopole position of pillow lavas is 032°E, 09°N (mainly normal polarities) and the position of the diabase dykes 202°E, 34°S (reversed polarities). The paleopole of the pillow lavas is obtained by combining the results relative to the paleopole position for pillow lavas from Part Ia (34°E @ 03°N) and for pillow lavas from Part II (30°E @ 14°N).

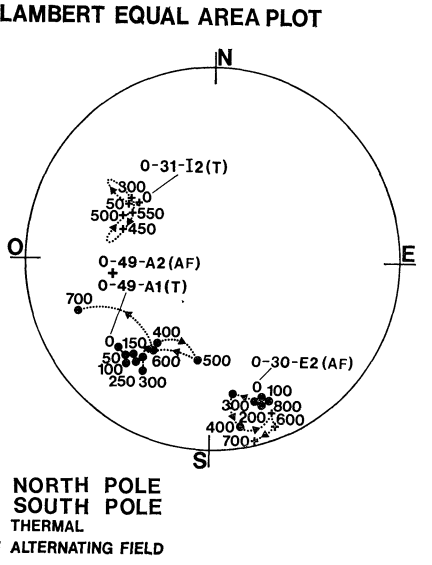


Fig. 3. Variation of the NRM vector directions of 4 pilot specimens of the ophiolite complex as a function of the AF demagnetizing strength and temperature.

8. Thermal Demagnetization

About 15% of the specimens were thermally demagnetized from 100 to 550°C by steps of 50°C. The rates of variation of the NRM direction and of the normalized intensity of 2 typical test specimens are presented in Figs. 1 and 2 respectively. The magnetization directions of 2 pilot specimens which underwent thermal demagnetization are presented in Fig. 3. The NRM orientations of the thermally demagnetized specimens coincide with the ones which were AF demagnetized in 40% of the cases only. In 40% of the cases, the NRM directions do not change appreciably in the thermal treatment which is apparently ineffective. A polarity inversion is commonly observed in the 350–500°C range. The median destructive temperature lies in the 250–400°C range except for those specimens depicting a polarity inversion and which have a blocking temperature in the 500–550°C range. Because the thermal treatment was only partially effective in the 300–550°C range, it was judged preferable to use AF demagnetization in order to obtain the paleomagnetic pole position.

9. Discussion of the Results

This paleomagnetic study was subdivided into two parts, i.e. Part I and II (Table 1). In Part I, up to nine samples of pillow lavas and diabase dykes were collected at each of the ten sites in order to find out if the paleopole position of the pillows is statistically different from that of the dykes. The second purpose of this comparative study was to find out if there is any statistical difference in the paleopole position of a relatively large number of samples collected on a few sites (Part I) and the paleopole position of a smaller number of samples collected on many sites (Part II). The answer to this last question is definitely no as can be seen directly from Table 1. However, the answer to the first question is not as straightforward.

Table 1 shows that the paleopole positions of the pillows (032°E, 09°N) and of the dykes once tilted (022°E, 34°N, normal) is closer to each other than before tilting correction, suggesting that the emplacement of the dykes is prefolding. However, the NRM polarity of the dykes is reversed while the one of the pillows is mixed (predominantly normal). Even after taking into account the polarity reversal, the paleopole position of the dykes and of the pillows is significantly different, this suggests either a relatively later time of emplacement of the dykes or the presence of an unproperly cleaned secondary NRM component in one or both lithological units.

Polarity reversals observed in the pillows were most commonly related to those specimens devoid of magnetite and containing sphene (grain size $\leq 15\ \mu$),

chromite ($\simeq 50\ \mu$), a vanadium-rich ferrosilicate containing two phases (grain size$\simeq 20\ \mu$, FeO$\simeq 80\%$) and native iron (grain size$\simeq 5\ \mu$, Fe$=98\%$) set in a chloritic gangue. All these ferromagnetic opaque minerals as well as chalcopyrite (5) were quantitatively analyzed with a scanning microprobe.

10. Paleopole Position

The paleopole position obtained in the ophiolites of Thetford Mines which are presumably of Middle Cambrian age does not fall on the Phanerozoic apparent wander track for North America (030°W, 16°S) (McELHINNY, 1973). This study suggests: 1) a short excursion of the north pole in the northern hemisphere during the Early to Middle Cambrian time or 2) a tectonic rotation of the ophiolitic plate or a fragment of it some 30–35° downward in the northwest direction around a horizontal axis oriented approximately N45°E and an anticlockwise rotation of the plate by an angle of approximately 25° about a vertical axis so that before it broke away from the original oceanic crust, this fragment was trending approximately N20°E. In spite of the fact that quite often ophiolite masses prove tectonically complex and internally unstable, it has been shown that certain aspects of the tectonic situation of the Thetford Mines ophiolites could be clarified by paleomagnetic means.

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